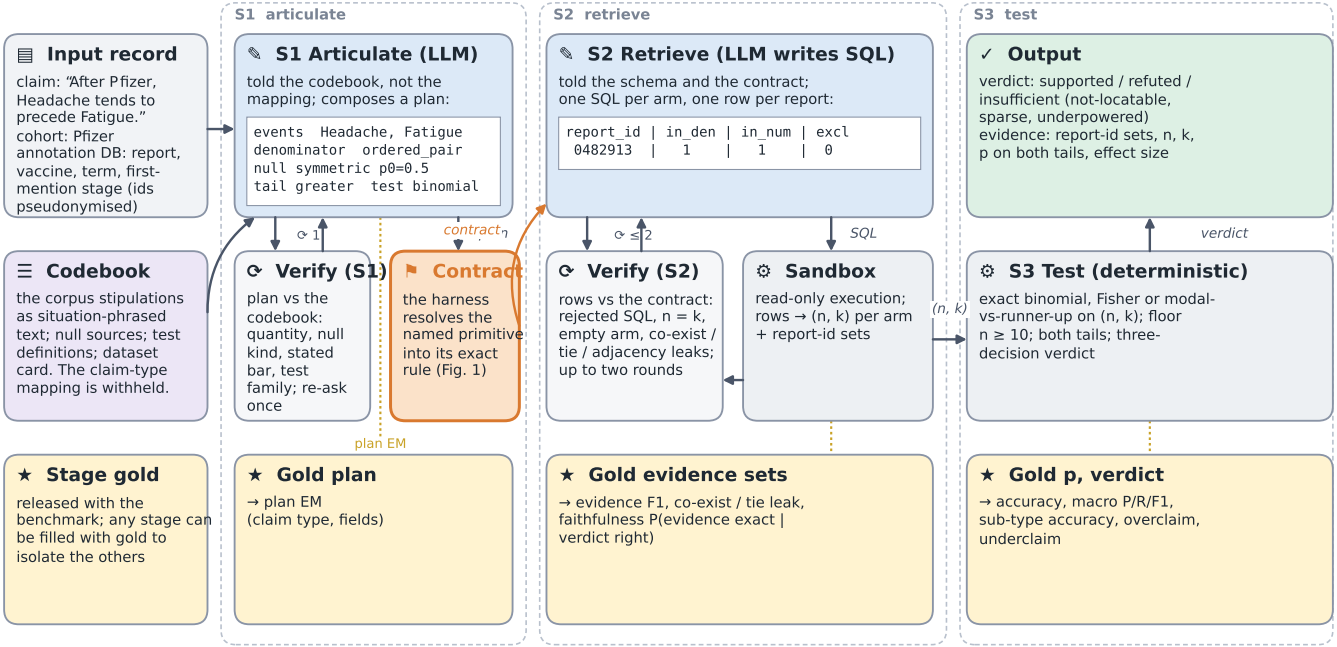

S1 articulate
S2 retrieve
S3 test
Input record
claim: "After Pfizer, Headache tends to precede Fatigue."
cohort: Pfizer
annotation DB: report, vaccine, term, first-mention stage (ids pseudonymised)
S1 Articulate (LLM)
told the codebook, not the mapping; composes a plan:
events Headache, Fatigue
denominator ordered_pair
null symmetric p0=0.5
tail greater test binomial
S2 Retrieve (LLM writes SQL)
told the schema and the contract; one SQL per arm, one row per report:
report_id | in_den | in_num | excl
0482913 | 1 | 1 | 0
Output
verdict: supported / refuted / insufficient (not-locatable, sparse, underpowered)
evidence: report-id sets, n, k, p on both tails, effect size
Codebook
the corpus stipulations as situation-phrased text; null sources; test definitions; dataset card. The claim-type mapping is withheld.
Verify (S1)
plan vs the codebook: quantity, null kind, stated bar, test family; re-ask once
⟳ 1
contract
Contract
the harness resolves the named primitive into its exact rule (Fig. 1)
plan EM
Verify (S2)
rows vs the contract: rejected SQL, n = k, empty arm, co-exist / tie / adjacency leaks; up to two rounds
⟳ ≤ 2
SQL
Sandbox
read-only execution; rows → (n, k) per arm + report-id sets
(n, k)
S3 Test (deterministic)
exact binomial, Fisher or modal-vs-runner-up on (n, k); floor n ≥ 10; both tails; three-decision verdict
verdict
Stage gold
released with the benchmark; any stage can be filled with gold to isolate the others
Gold plan
→ plan EM (claim type, fields)
Gold evidence sets
→ evidence F1, co-exist / tie leak, faithfulness P(evidence exact | verdict right)
Gold p, verdict
→ accuracy, macro P/R/F1, sub-type accuracy, overclaim, underclaim